

Design Pattern 8: Strategy Pattern

Exercise 8: Implementing the Strategy Pattern

Scenario:

You are developing a payment system where different payment methods (e.g., Credit Card, PayPal) can be selected at runtime. Use the Strategy Pattern to achieve this.

Steps:

Create a New Java Project:

Create a new Java project named StrategyPatternExample.

Define Strategy Interface:

Create an interface PaymentStrategy with a method pay().

Implement Concrete Strategies:

Create classes CreditCardPayment, PayPalPayment that implement PaymentStrategy.

Implement Context Class:

Create a class PaymentContext that holds a reference to PaymentStrategy and a method to execute the strategy.

Test the Strategy Implementation:

Create a test class to demonstrate selecting and using different payment strategies.